8) Experiment to understand Linear regression for the given data set.

Program:

```
import numpy as np
import pandas as pd
df = pd.read_csv('/Salary_data.xlsx - Salary_data.csv')
df
df.info()
df.dropna(inplace=True)
df.info()
df.describe()
features = df.iloc[:, [0]].values
label = df.iloc[:, [1]].values
from sklearn.model_selection import train_test_split
x_train, x_test, y_train, y_test = train_test_split(
    features, label, test_size=0.2, random_state=0
)
from sklearn.linear_model import LinearRegression
model = LinearRegression()
model.fit(x_train, y_train)
model.score(x_train, y_train)
model.score(x_test, y_test)
model.coef_
model.intercept_
import pickle
pickle.dump(model, open('SalaryPred.model', 'wb'))
model = pickle.load(open('SalaryPred.model', 'rb'))
yr_of_exp = float(input("Enter Years of Experience: "))
yr_of_exp_NP = np.array([[yr_of_exp]])
```

```
Salary = model.predict(yr_of_exp_NP)
print("Estimated Salary for {} years of experience is {}: ".format(yr_of_exp, Salary))
```

Output:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   YearsExperience  30 non-null     float64
 1   Salary           30 non-null     int64
dtypes: float64(1), int64(1)
memory usage: 612.0 bytes
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   YearsExperience  30 non-null     float64
 1   Salary           30 non-null     int64
dtypes: float64(1), int64(1)
memory usage: 612.0 bytes
Enter Years of Experience: 10.4
Estimated Salary for 10.4 years of experience is [[123630.88046861]]:
```

Result:

Experiment is conducted to understand linear regression of the given data set.